Tutorial Questions: Calculations in Fraud and Detection Technologies

Section A: Percentages and Rates

1. A fraud detection system identified 72 fraudulent transactions out of 90 actual fraud cases.
Calculate the fraud detection rate percentage.
2. A bank processed 5,000 transactions in one day, and 125 were flagged as suspicious.
Calculate the percentage of suspicious transactions.
3. Out of 200 flagged transactions, 150 were confirmed as actual fraud cases.
Calculate the fraud accuracy rate.
4. A company lost ZMW 45,000 due to fraud last year and ZMW 30,000 this year.
Calculate the percentage decrease in fraud losses.
5. An organization recovered ZMW 18,000 from a total fraud loss of ZMW 24,000.
Calculate the recovery rate percentage.

Section B: Average and Mean Calculations

6. The number of fraud cases detected over five months were:
12, 18, 20, 15, and 25.
Calculate the average number of fraud cases detected per month.
7. A fraud analyst recorded the following transaction monitoring times (in minutes):
5, 8, 6, 10, 7, 9.
Find the mean monitoring time.
8. The financial losses (in ZMW) from four fraud incidents were:
2,500; 3,000; 4,500; and 5,000.
Calculate the average financial loss.

Section C: Variance and Standard Deviation

9. The number of fraudulent login attempts detected over five days were:
4, 6, 8, 5, 7.
Calculate the variance.
10. A fraud monitoring system recorded the following suspicious transactions per hour:
10, 12, 9, 11, 8.
Calculate the standard deviation.
11. A cybersecurity team recorded these phishing attack reports per week:
15, 18, 20, 22, 25.
Calculate the variance and standard deviation.

Section D: Probability and Risk Analysis

12. In a database of 1,000 transactions, 50 were fraudulent.
What is the probability that a randomly selected transaction is fraudulent?
13. A fraud detection tool correctly identifies fraud 85 times out of 100 attempts.
Calculate the probability of successful fraud detection.
14. Out of 300 employee accounts, 12 were compromised.
Calculate the probability of selecting a compromised account.

Section E: Ratios and Comparisons

15. A company recorded 40 online fraud cases and 10 internal fraud cases.
Express the ratio of online fraud to internal fraud.
16. A fraud department solved 24 fraud cases out of 30 reported cases.
Express the solved-to-reported fraud ratio in simplest form.

17. A financial institution spent ZMW 120,000 on cybersecurity and ZMW 80,000 on fraud investigations.
Find the ratio of cybersecurity spending to fraud investigation spending.
Section F: Advanced Application Questions
18. A fraud detection algorithm analyzed 2,000 transactions and flagged 160 as suspicious. Later, 120 were confirmed fraudulent.
Calculate:
The suspicious transaction percentage
The fraud confirmation rate
19. The fraud losses (in ZMW) for six branches were:
5,000; 6,500; 7,000; 8,000; 4,500; 9,000.
Calculate:
Mean
Variance
20. A digital payment platform experienced the following fraud cases over four quarters:
Q1 = 15, Q2 = 20, Q3 = 18, Q4 = 27
Calculate:
Average fraud cases
Percentage increase from Q1 to Q4
21. A fraud analyst reviewed 250 transactions in a day and detected 35 fraudulent transactions.
Calculate the fraud detection percentage.
22. A machine learning fraud model predicted 90 fraud cases, but only 75 were actual fraud cases.
Calculate the prediction accuracy percentage.
23. A cybercrime unit recovered ZMW 60,000 from total losses of ZMW 100,000.
Calculate the percentage recovered.
24. A fraud monitoring center recorded these daily fraud alerts:
14, 16, 13, 18, 19, 20, 15
Calculate:
Mean
Variance
Standard deviation
25. A bank had 25 fraud incidents among 2,500 customers.
Calculate the fraud incident probability and express it as a percentage.